

Where to Stop: Protocol Audits and the Next Controlled Experiments

Anonymous CVPR submission

Paper ID DRAFT

S1. Scope and Reproduction

This supplement records CPU analyses of the frozen evaluation archive and a prospective protocol for the ongoing depth study. The main manuscript uses the recorded mixed reconstructions of the shared-latent e15 codec. The new control-path and spatial-assignment analyses reuse its stored source-error tables. They produce candidate decisions for later replay, without executing the codec or measuring new inference times. The independent D2/D4/D6 training continues separately; no intermediate checkpoint is treated as a completed depth experiment.

The stand-alone paper repository contains the plotted source tables, source hash manifests, editable figure scripts and inspection results under `data/refresh20260927`. Figures can be regenerated on CPU using `scripts/build_figures.py` and `scripts/build_extended_figures.py`. The parent research repository also retains the scripts that derive these summaries from the raw records and checkpoints. Hashes identify the files inspected during this revision; they do not retrospectively establish the exact executable used by a historical run whose code hash was not recorded.

S2. Tracing the Reconstruction Reference

Two measurement paths need to be distinguished. The source-table producer loads the e15 network and a separate released-weight reference. It computes the candidate tile errors $M_{t,k}$ from e15 and the scalar reference error R from the released synthesis network, both on the padded source. The final-reconstruction evaluator instead loads e15 alone, calls its full-frame synthesis path, crops both full and mixed outputs, and measures their difference in source quality. Its raw field named `psnr_release` is consequently a fine-tuned full-depth reference in the inspected code (Fig. S1).

A CPU comparison of the two state dictionaries finds 255 shared encoder/entropy tensors with no changes. All 143 inherited synthesis tensors that can be compared have changed, with a maximum absolute parameter difference of 0.2481. Thus the reporting path cannot be identified with released synthesis merely because it executes all twelve trunk

blocks.

This distinction limits the interpretation of threshold crossings. The recorded 7/263 RGB and 49/263 YUV losses above a nominal 0.1 dB target combine differences in reference, image support, metric and the influence of the shared repair/head. They do not identify the contribution of any single factor. The paired MAC accounting remains attached to the same recorded maps, but the exact quality gap to the cropped released decoder cannot be recovered from the exported JSON alone.

The required replay should retain both full-depth anchors, the padded and valid-pixel errors, and the final mixed reconstruction. Selection and final verification must use the same declared reference and metric. A failed plan should trigger an explicit fallback whose bytes and runtime are recorded.

S3. Auditing the Scalar Control Search

For router scores $\ell_{t,k}$ and costs c_k , increasing the price β in $k_t(\beta) = \arg \max_k (\ell_{t,k} - \beta c_k)$ cannot increase selected cost. To see this, write the optimality inequalities at $\beta_1 < \beta_2$ for their respective choices k_1, k_2 and add them. The result is

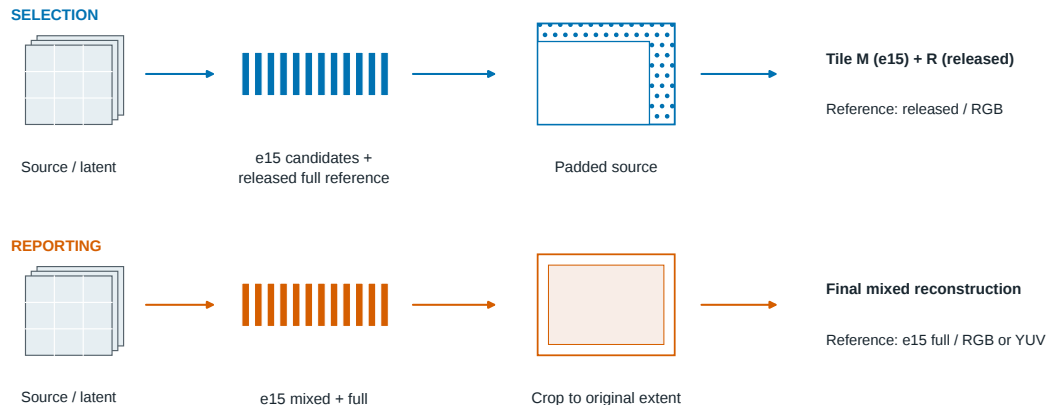
$$(\beta_2 - \beta_1)(c_{k_1} - c_{k_2}) \geq 0. \quad (\text{S1})$$

No corresponding statement follows for source error $M_{t,k}$, because that quantity is absent from the score being optimised. A feasibility test at the most expensive endpoint can therefore miss a feasible middle assignment, and distortion-based bisection can miss a later feasible interval.

We enumerate pairwise score-line intersections, representatives of the open intervals between them, and the endpoints of the archived range $[-20000, 20000]$. Archived controls are included explicitly to retain their floating-point tie behaviour. Identical maps are merged. Among feasible maps, the selected candidate maximises modelled MAC saving, with table loss as a tie-breaker. This is a finite-path audit in float64, rather than an optimiser over all possible tile assignments.

The cost is monotone on every inspected router path, while source loss has at least one reversal on 227/265 paths. At 0.05 dB, enumeration improves 3 of 204 previously feasible plans, by 0.1626 percentage points on average, and

a Selection and reporting use different reconstruction anchors and image support



CPU audit: 255 shared encoder/entropy tensors unchanged; 143/143 inherited synthesis tensors changed.

Required comparison: reconstruct both anchors, use the same valid pixels and metric, then verify the selected output.

Figure S1. **Reference and support are part of the measurement.** Candidate tile errors come from e15, whereas the table denominator comes from a separate released-weight reference. Final reporting uses the cropped full-depth e15 reconstruction. The CPU tensor comparison establishes a weight difference in the inspected checkpoints. The figure traces code paths and does not depict newly decoded images.

recovers one additional feasible case. At 0.1 dB it improves 2 of 263 plans, by 0.0277 points on average. No paired saving changes at the four larger targets. The largest missed saving is 30.45 points at 0.05 dB, so a small mean difference should not be read as a uniform per-frame bound.

We strengthen the blind baseline in the same way. For ordered dithering, all changes occur at a finite set of Bayer-rank thresholds and rung boundaries. Enumerating those maps reveals source-loss reversals on 232/265 paths. It improves 3/204 paired plans at 0.05 dB and 1/263 at 0.1 dB, with mean gains of 0.0192 and 0.0028 points, respectively. Six additional cases become feasible at 0.05 dB. The quality of these newly selected maps has not been measured after final mixed reconstruction, so they do not replace the main manuscript’s reconstruction curves.

The audit reproduces every archived feasible control and cost, verifies cost monotonicity, and checks that enumeration never discards a better archived feasible saving. An independent three-exit example verifies the non-monotone failure mechanism against a dense control grid.

S4. Separating the Histogram from Spatial Placement

A map can save arithmetic by selecting many cheap exits even if their locations contain no useful information. To isolate placement, let $p_k = N^{-1} \sum_t \mathbf{1}[k_t = k]$ be its fixed exit histogram and let π be a uniformly random permutation of its assignments. Each tile has marginal exit probability p_k , giving the exact expectation

$$\mathbb{E}_\pi \left[\frac{1}{N} \sum_t M_{t, k_{\pi(t)}} \right] = \frac{1}{N} \sum_t \sum_k p_k M_{t, k}. \quad (\text{S2})$$

We report $G = 10 \log_{10}(\mathbb{E}_\pi[\overline{M}_\pi]/\overline{M})$, where $\overline{M}$ is the recorded map’s table MSE. Positive G indicates useful spatial placement at exactly the same histogram and MAC cost. The logarithm is outside the expectation: this is not expected PSNR over random reconstructions.

At 0.1 dB, source-informed search has a mean placement gain of 0.0321 dB, the router 0.0213 dB, and dithering 0.0007 dB over their respective shuffled histograms. The router’s 95% sequence-cluster interval is [0.0167, 0.0262] dB; dithering’s is [−0.0002, 0.0017] dB. Uniform-depth maps have zero gain by construction. Among the 263 paired router maps, 204 have positive gain, seven have negative gain and

52 use one exit only. This diagnoses content association conditional on a source-calibrated histogram; it does not validate held-out autonomous calibration.

For each map we also solve the exact fixed-histogram assignment problem on M . Repeating each exit column according to its count produces a square linear assignment problem. Its minimum table MSE gives a conditional upper bound on placement value. The router’s mean regret to that optimum is 0.0044 dB at the 0.1 dB target. Every source-search map attains its own fixed-histogram optimum, as expected for a separable Lagrangian minimiser. The optimum still requires the complete source table and carries the same information-acquisition cost. Replaying shuffled and optimal maps through the mixed decoder is the next required ablation.

S5. The Independent Depth Study

The ongoing experiment retains the first two, four or six synthesis trunk blocks of the DCVC-UF intra architecture and trains each complete codec under the pinned official recipe. The encoder and entropy weights learn alongside the decoder. Each model consequently has its own representation, distinct from the shared-latent exits evaluated in the main paper. Figure S3 shows a possible later deployment path, not an implemented six-expert result.

Isolating depth. Released D12 is a useful deployment anchor, but a D12 trained with the same data manifest, 105-epoch schedule, optimisation and initialisation protocol is necessary to isolate depth from training history. D8 and D10 should be prioritised after the initial depth frontier and this control are available. The main runs retain the official recipe; distillation and prefix initialisation are separate variants rather than silent training changes.

Isolating routing. The evaluation should progress from full-frame D12 to patchwise D12, the best validation-selected fixed expert, blind mixtures, simple image statistics, the MLP and source-informed search. Full-frame versus patchwise D12 measures the cost of tiling and entropy resets before routing receives credit for any gain. A histogram-preserving shuffle separates the chosen expert proportions from their spatial placement.

A concrete router hypothesis. A lightweight encoder-side predictor can estimate each expert’s distortion and rate relative to a cheap reference. Given measured runtime tables, a candidate decision score is

$$\hat{J}_k = \hat{D}_k + \lambda \hat{R}_k + \mu T_k(n_k, \text{shape}). \quad (\text{S3})$$

Here n_k is the number of patches assigned to expert k ; grouped execution can make runtime depend on the assign-

ment as a whole. The score is therefore a proposed approximation requiring measured validation, rather than an independent per-patch latency oracle. Regression or regret-weighted training should be compared against ordinary exit-label cross-entropy. A second-stage top-two verification can be restricted to small predicted margins, with its threshold selected on held-out calibration images and its extra encoding work included in the total.

Spatial competition between complete codecs already provides a close precedent [1]; adaptive compression and patch early exit provide further controls [2, 4, 6]. The testable direction here is depth-heterogeneous selection with measured rate–distortion–time costs and reduced encoder search. Practical codec studies motivate including device execution and perceptual assessment [3, 5].

S6. Evaluation Contract for the Next Stage

Expert training, router training, calibration and final evaluation should be split at image level. Patches from the same image must stay in one split. The four fixed validation crops in the current training monitor are health checks; a broader intermediate evaluation should use the available 100 DIV2K validation images, followed by the declared final Kodak/CTC/CLIC scope. This expanded evaluation is proposed and has not been completed.

Each codec needs a true-byte rate–distortion curve. Equal QP does not imply equal rate across independently learned codecs. Comparisons should use their common measured support, reporting infeasible targets and avoiding extrapolation. Rate includes main and hyperprior payloads, model identity, quality/control parameters and framing headers, divided by original pixels. Quality is measured after cropping valid pixels; the exact RGB/YUV formula and perceptual metrics are declared before test-set selection.

Timing requires the same GPU, precision, kernel path and warmup, with alternated paired execution order and retained raw trials. Encoder, decoder and complete round-trip times are separate outcomes. CUDA-event module time and end-to-end wall time must remain separately labelled. Grouping, scatter/gather, entropy work, stream parsing, transfers and fallback are included where they occur. The full codec bank’s resident weight memory and peak activation memory are both reported.

The primary success criterion is a positive net runtime advantage over the best validation-selected blind control on common achieved rate–distortion support. Critical comparisons should then be repeated across training seeds. A content bootstrap at fixed weights measures a different source of variation and cannot replace those repeats.

References

[1] Théophile Blard, Pierrick Philippe, Théo Ladune, Xiaoran Jiang, and Olivier Déforges. Spatial competition for

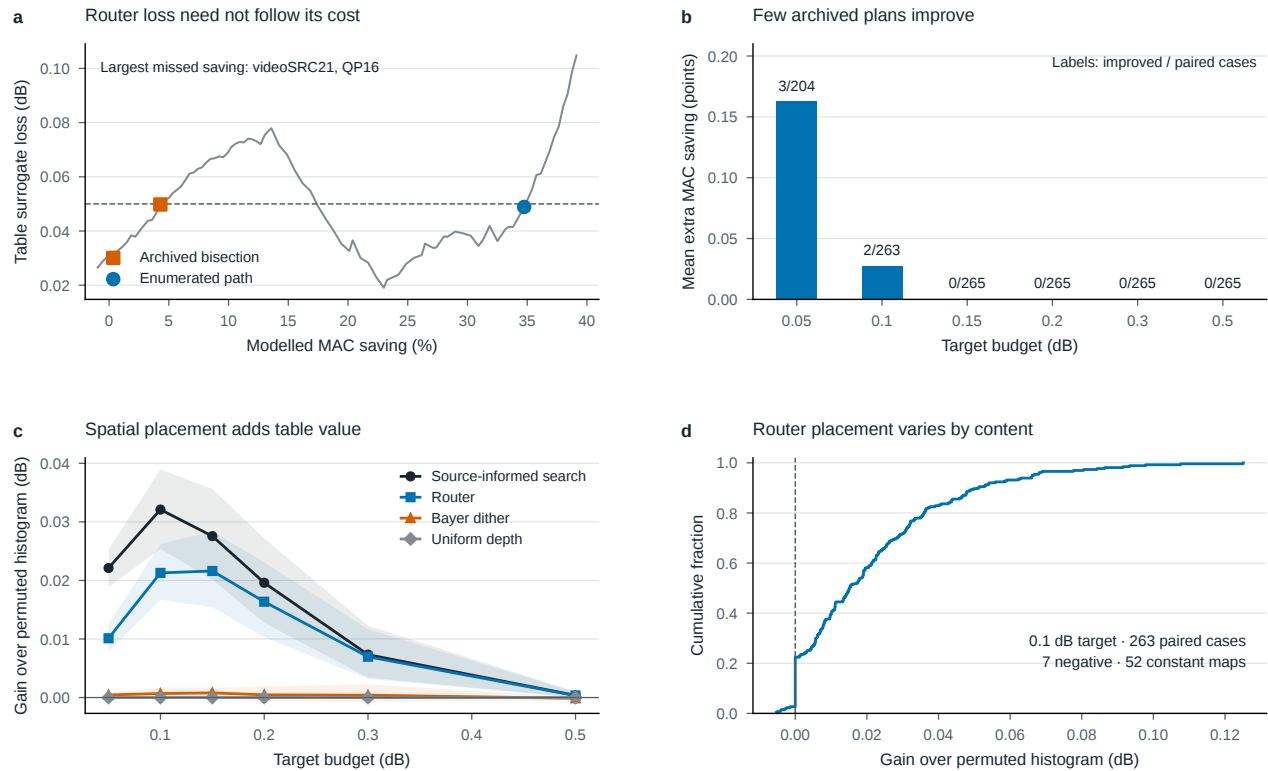
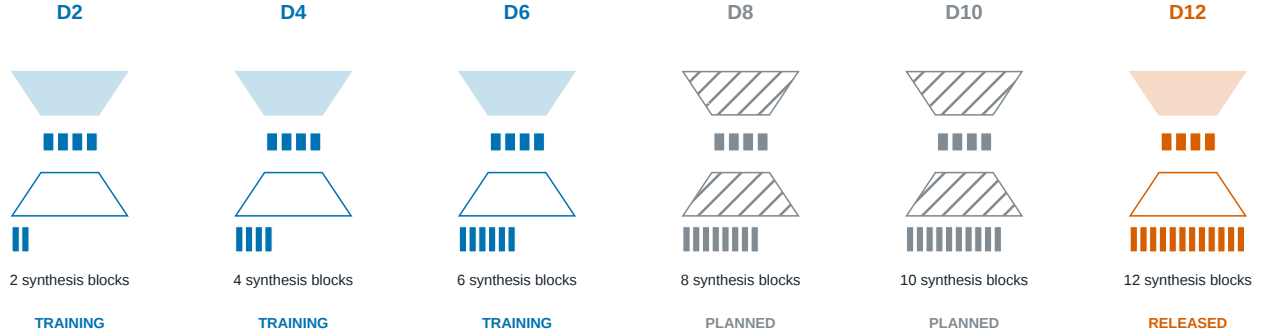


Figure S2. **Control-path and assignment diagnostics.** **a**, The explicitly selected largest missed router saving illustrates non-monotone table loss. **b**, Mean paired improvement from enumerating the router path; labels show improved and paired counts. Newly feasible cases are excluded from these means. **c**, Exact expected-shuffle comparison at unchanged exit histograms, with 95% sequence-cluster intervals. **d**, Router placement-gain distribution at 0.1 dB. All results are CPU analyses of padded error tables; none measures new final-image quality or latency.

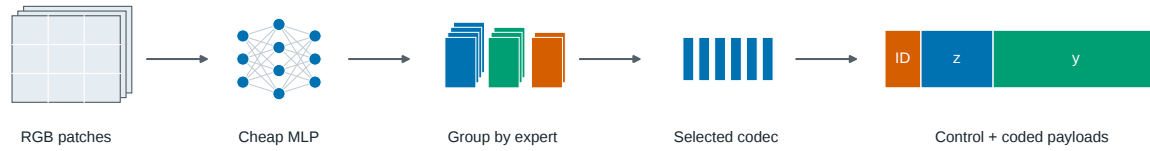
- low-complexity learned image compression. *arXiv preprint arXiv:2605.13243*, 2026. 3
- [2] Zhihao Hu and Dong Xu. Complexity-guided slimmable decoder for efficient deep video compression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14358–14367, 2023. 3
- [3] Zhaoyang Jia, Bin Li, Jiahao Li, Wenxuan Xie, Linfeng Qi, Houqiang Li, and Yan Lu. Towards Practical Real-Time Neural Video Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3
- [4] Lvfang Tao, Wei Gao, Ge Li, and Chenhao Zhang. Adanic: Towards practical neural image compression via dynamic transform routing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16879–16888, 2023. 3
- [5] Kedar Tatwawadi, Parisa Rahimzadeh, Zhanghao Sun, Zhiqi Chen, Ziyun Yang, Sanjay Nair, Divija Hasteer, and Oren Rippel. What matters in practical learned image compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 3
- [6] Shizun Wang, Jiaming Liu, Kaixin Chen, Xiaoqi Li, Ming Lu, and Yandong Guo. Adaptive Patch Exiting for Scalable

Single Image Super-Resolution. In *European Conference on Computer Vision (ECCV)*, 2022. 3

a Independent codecs: depth changes the synthesis network, training changes the whole codec

D2 / D4 / D6: pinned official recipe, all codec weights learned jointly

A same-recipe D12 control is still needed to separate depth from training provenance.

b Proposed deployment: choose before encoding, group work by expert, transmit the choice

The decoder reads the expert ID and uses that expert's entropy model and synthesis transform.

Measure: full bytes · encoder + decoder wall time · final cropped quality · bank memory

Deployment is a proposed design. The glyphs and queues are schematic; they do not show measured routes or throughput.

Figure S3. **An independent codec bank is a separate experimental system.** Training status is shown as of 27 September 2026. D2/D4/D6 are in progress, D8/D10 are planned, and D12 is the released reference. Each expert owns its analysis, entropy and synthesis parameters. The proposed encoder-side MLP selects an expert before encoding; the decoder receives the expert ID. Queues, network shapes and stream segments are schematic. No throughput or quality result is encoded by their geometry.